

Power Atlantic 5MWh Battery Container

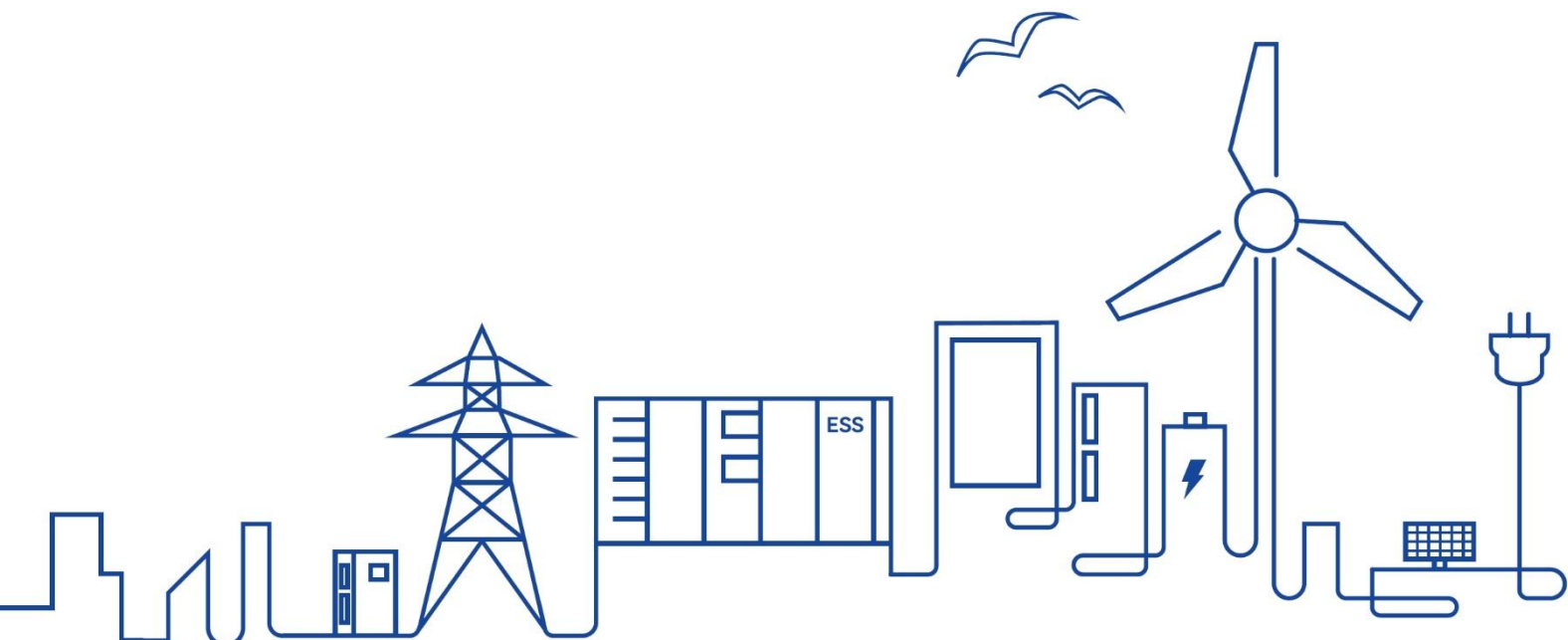
Maintenance Manual

(Preliminary Version)

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February 2025



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1 Routine Maintenance

1.1 Precautions Before Maintenance

 WARNING	● Do not open the door to maintain the device in rainy, humid or windy days. LINYANG shall not be held liable for any damage caused by violation of the warning. ● Avoid opening the container door when the humidity is high in rain, snow or fog, and make sure that the seals around the container door do not curl when the door is closed. ● Before performing maintenance on a Rack, make sure that all MSDs in that Rack have been properly disconnected. ● When performing maintenance on the Pack, ensure that the locking structure of the plug is in the unlocked state before unplugging the power cord to avoid damage to the Pack caused by strong unplugging.
 WARNING	● To avoid electric shock, do not perform any other maintenance operations beyond this manual. ● If necessary, contact LINYANG customer service for maintenance.
 NOTICE	● In the event of heavy snowfall at the project site, please clear the snow from the top of the equipment and the surrounding area in a timely manner.

*In fair weather, it is recommended to open the container door to dehumidify the equipment.

1.2 Maintenance Item and Interval

* This section is the recommended maintenance cycle. The actual maintenance cycle should be adjusted according to the specific installation environment of this product.

*The power station scale, installation location and on-site environment affect the maintenance cycle of this product. In sandy or dusty environments, it is necessary to shorten the maintenance cycle and increase the frequency of maintenance.

1.2.1 Maintenance (Initial grid connection)

Check the following items and correct them immediately if they do not meet the requirements:

- Examine the input and output cable materials and specifications.

- Terminal materials, specifications and their installation direction.
- Bolt size and its gasket mounting direction.

1.2.2 Maintenance (Every month)

Item	Check Method
Container	● Check whether there is any damage, flaking paint or sign of oxidization on the enclosure. ● Check whether there is any damage or deformation of the container and internal devices. ● Check whether there are flammable objects on the top of the container. ● Check whether the welding points between the container and the foundation steel plate are firm and whether there is corrosion. ● Check whether the lock of the container door can be unlocked flexibly. ● Check whether the sealing strip is fixed properly. ● Check whether there are foreign objects, dust, dirt, and condensed water inside the integrated energy storage system.
Air inlet and outlet Cables	● Check to see if the outdoor container air inlet and outlet are blocked. ● Check all cables for breakage.
System status	● Check if there is abnormal noise during operation of internal devices. ● Check whether the temperature in the container is excessively high. ● Check whether the humidity and the amount of dust inside the container are within the normal range. Clean the equipment if necessary.
Software data	● Read the data from BMS ● Save run data, parameters, and logs to the relevant files. ● Check the various parameter settings. ● Update the software.

1.2.3 Maintenance (Every half a year)

Item	Check Method
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Safety function	● Check whether the emergency stop button work normally. ● Simulate shutdown. ● Check the warning marks and other device marks, and replace them timely when they are fuzzy or damaged.
Internal components inspection	● Check the cleanness of the circuit board and other elements and components. ● Check to see if the fan is operating normally and if there is any abnormal noise. ● Check the temperature of the radiator and the amount of dust accumulated. Clean heat-dissipation modules with a vacuum cleaner if necessary. ● Replace the air filter screen when necessary.
Device maintenance	● Carry out regular inspection for corrosion of all metal components (once per half a year). ● Check the contactors (auxiliary switches and micro-switches) annually to ensure the good mechanical operation. ● Check the running parameters (especially voltage and insulation). ● Troubleshoot for unoperated UPSs, which need to be recharged once every six months.

1.2.4 Maintenance (Once a year)

Item	Check Method
Ground of the shielded layer of cables	● Check whether the cable shielding layer is in good contact with the insulation sleeve and whether the copper bus bar is firmly fixed.
Surge protection device and fuse	● Check whether the SPD and fuse are properly fastened.

Wiring and cable layout	● Check whether the cable layout is normal and whether there is a short circuit. For any non-conformances found during inspection, correct them immediately. ● Check whether all cable entry are well sealed. ● Check whether the power cables are loose, and fasten them again by the torque specified previously. ● Check whether the power cables and control cables are damaged, especially if the surface contacting the metal surface is cut. ● Check whether the insulation tapes on the power cable terminals fall off.
Ground connection and equipotential Connection	● Check whether the ground connection is correct and the grounding resistance shall be no more than 4Ω. ● Check whether the equipotential connection inside the integrated system is correct.